Emotion Classification of Lyrics through Summarization by Large Language Models

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***Abstract*—We propose a method that utilizes a large language model in the task of lyrics emotion classification. We especially employ GPT-4o, which is expected to deliver high performance to conduct emotion classification of lyrics, where we propose a few-shot prompt of GPT-4o for classification into 6 classes. We developed a dataset of 181 lyrics categorized into six classifications to evaluate GPT-4O's performance. This dataset maintains a reasonable level of validity as it was carefully curated by the first author, then independently reclassified by six annotators, with final classifications determined through majority voting. Performing detailed classification, we achieved over 75% classification performance for the total accuracy. This accuracy was achieved through performing extractive summarization of the lyrics into optimal number of characters. In other words, rather than feeding the collected lyrics directly into GPT-4o, we implemented a two-stage process where GPT-4o first automatically summarizes the lyrics before they are used. Using a confusion matrix, we analyzed the tendencies of classification errors and showed the possibility of improving the total accuracy.**

***Index Terms*—lyrics emotion classification, large language models, text classification**

I. Introduction

In recent years, music websites and applications offer playlists that match specific emotions when searching for music based on those emotions. These playlists are manually compiled by the operators of the websites or applications, or by users. Music recommendation systems [1], [2] have been studied, and the demand for emotion classification of music is increasing. However, music consists of several elements, and it is sometimes unclear which criteria users who compile the playlists prioritize, such as tempo, rhythm, tone, or lyrics, when classifying emotions. Thus, the impression of music is considered to be highly subjective. But, when focusing solely on lyrics, one realizes that there are roles that only lyrics can fulfill. For example, whether a song is about friendship or love is something that can be understood by reading the lyrics. Therefore, in this paper, we have decided to focus on lyrics, which can categorize music into detailed classifications.

Here, a method of using techniques such as machine learning to classify lyrics can be considered. However, currently, there is no large-scale dataset of Japanese lyrics that is manually annotated with emotion, which is inevitable for training machine learning models for lyrics emotion classification. Therefore, there exists a challenge in classifying the emotions of lyrics due to the lack of training data. To address this issue, methods have been proposed to automate the emotion classification of lyrics and generate classified data, such as a method [3] that matches lyrics with a dictionary database and performs lyrics emotion classification using a unique scoring method.

In contrast, this paper uses a large language model (hereinafter abbreviated as "LLM") to perform emotion classification of lyrics. In this paper, we adopt GPT-4o[1], which is considered to have particularly excellent performance, as an LLM and apply it to the task of emotion classification of lyrics. We especially propose a few-shot prompt of for classifications into 6 classes. As shown in Fig 1(a), to verify the performance of GPT-4o in detailed emotion classification, we set up classification into 6 emotions. Here, lyrics typically contain a variety of stories, and a single lyric can often encompass multiple emotional classifications. As a result, it is common to assign multiple classes or to specify emotions as components. In this study, however, we decided to determine and assign a single class that closely approximates the overall emotion. The advantage of this approach is that it becomes easier to judge whether the class assigned to the lyrics is correct or not. For instance, with multi-class assignment or component specification, correct and incorrect classes or components often coexist, leading to ambiguous judgments like "it might be somewhat correct" or "it might be somewhat incorrect." Therefore, we opted for assigning only one class, as it makes it easier to judge the correctness of that single class. As a result, we achieved the total accuracy of 77%. We studied the confusion matrices of classification results , where we observed frequent classification errors and investigated future directions for further improvement.

Moreover, in this paper, the emotion classification performance has been improved by performing manual extraction of the lyrics into optimal length. However, the problem is that this task is time-consuming. Therefore, we also had GPT-4o perform the task of extractive summarization of lyrics into optimal length and evaluated the performance of emotion classification using the resulting lyric data.

Overall, the contributions of this paper are summarized as below:

1) In order to address the issue of the lack of emotion classification datasets for Japanese lyrics, this paper

[1]https://openai.com/index/gpt-4/

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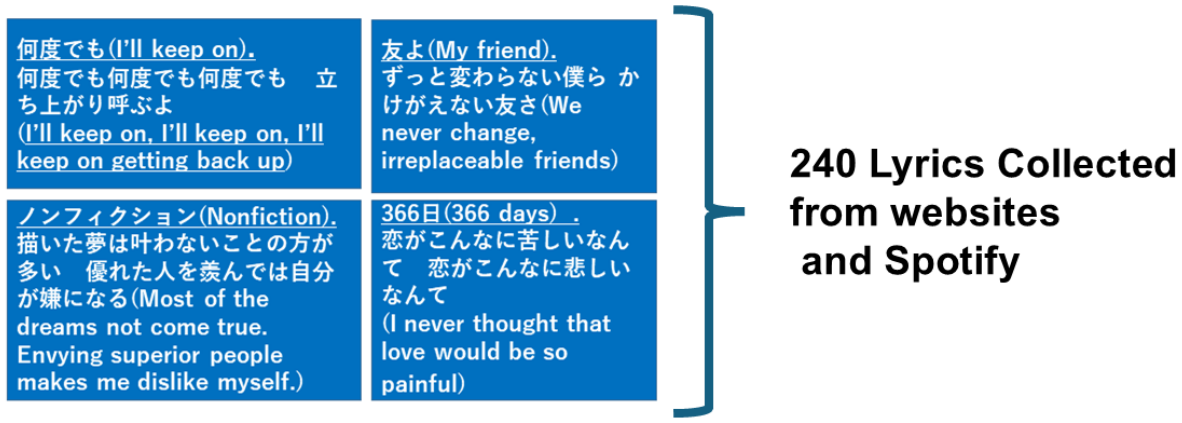
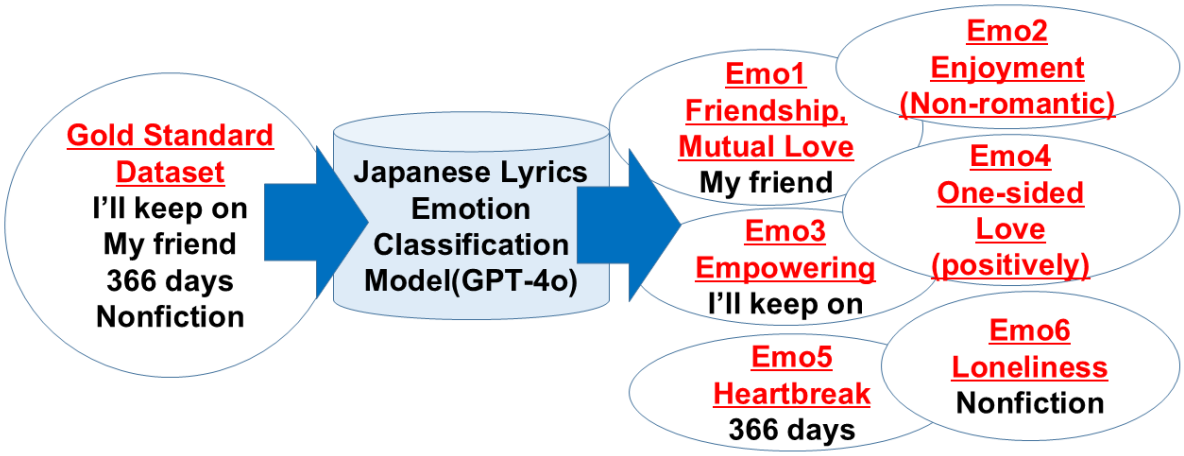


(a) Examples of Collected Lyrics



(b) Overview of Lyrics Emotion Classification

Fig. 1: Overall Lyrics Emotion Classification (Song title, artist, lyric writer, and composer for the 4 songs are: title: "I'll keep on", artist: DREAMS COME TRUE, writer and composer: Miwa Yoshida, Masato Nakamura, title: "366days", artist: HY, writer and composer: Izumi Nakasone, title: "Nonfiction", artist: Ken Hirai, writer and composer: Ken Hirai, title: "My friend", artist: Ketsumeishi, writer and Composer:Ketsumeishi, Naoyuki Honzawa,T omoyuki Tajiri)

studies how to apply an LLM to the task of Japanese lyrics emotion classification.

2) We developed a Japanese lyrics emotion classification dataset consisting of 240 lyrics with classification into 6 classes.
3) We propose a few-shot prompt of GPT-4o for classifications into 6 classes, achieving over 73% total accuracy
4) We were able to improve the emotion classification performance of lyrics by more than 10% total accuracy by having GPT-4o perform the task of extractive summarization of lyrics compared to when summarization of lyrics was not performed.

II. Related Work

Related studies to this paper include previous research on emotion classification of songs or lyrics. For example, Malheiro et al. [4] mapped emotion words assigned to each song on the music website AllMusic[2] into a two-dimensional space. Using this, they created a classification dataset of 400 songs from each quadrant (a total of 4 quadrants) of the two-dimensional space. They performed manual judgment of emotion classification and created and published 771 cases of song emotion classification. Cano et al. [5] used three dictionaries to score lyrics with two values, Vt (valence) and At (arousal), and mapped lyrics into a two-dimensional space. They created and published a dataset of 2,595 lyrics classified into 4 moods. While these studies allow for searching for lyrics that match emotions, however, they do not address the issue of searching for lyrics that match situations.

Revathy et al. [6] employed the MER dataset [7], in which

[2]https://www.allmusic.com/

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TABLE I: Detailed Description of Emotion Classes

ID	Class	Detailed Description	# lyrics
⟨Emo1⟩	friendship mutual love	expressing caring thoughts towards friends or expressing mutual affection for partners or family	30
⟨Emo2⟩	enjoyment (non-romantic)	many joyful phrases expressing happiness through actions	26
⟨Emo3⟩	empowering	encouraging and inspirational words for oneself or others	36
⟨Emo4⟩	one-sided love (positively)	expressing one-sided romantic feelings	26
⟨Emo5⟩	heartbreak	expressing deep sadness over a breakup or ended relationship with partner	28
⟨Emo6⟩	loneliness	expressing longing to fill an emotional void but feeling unfulfilled	35
Total			181

English lyrics are classified into 4 emotions, and performed transfer learning from the MER dataset to the Music4all dataset [8] that had not been emotion classified.

Research on music search models using lyrics is also being conducted in Japan [9], [10]. Tsukuda et al [11]. constructed a topic model using lyrics data from 147,990 songs by 3,722 artists, and were able to assign multiple topics to each set of lyrics, such as 'sentimental' and 'devoted love'. However, the dataset used is restricted internally and cannot be freely used. In other words, lyrics data that anyone can reference is not yet publicly available. In this regard, an advantage of this research is that it uses LLM for emotional classification of lyrics, which does not require large-scale training data.

Compared with those previous works, this paper aims to construct a Japanese lyrics dataset annotated with emotion classification. Here, in this paper, the main focus is on utilizing an LLM with this task, which have never been studied so far in the context of lyrics emotion classification.

III. Lyric Data Preparation

This section describes the data preparation method as well as the classes and the criteria for emotion classification. Table I shows the number of lyrics collected in this paper. A total of 181 lyrics were collected, where all of these were used for the evaluation of emotion classification.

A. Lyrics Collection and Classification

First, we needed to establish appropriate categories for lyrics classification. Table I shows the details of the emotion classes. In the initial stage of this study, lyrics were classified based on emotion words, referring to Russell's circumplex model [12]. However, we concluded that lyrics contain not only a single emotion but also multiple emotions, and cannot be fully expressed by a single emotion word. Therefore, as shown in Table I, we set up 6 unique emotion classes which is incorporating situation-specific categories like 'heartbreak' and 'friendship/mutual love'. As shown in the Table I, detailed descriptions are provided for each category. We carefully reviewed the lyrics and collected relevant examples by referring to those criteria. Thus, we collected 120 lyrics from websites and 120 lyrics from Spotify [3] , with 40 lyrics for each category. However, this would result in a highly subjective dataset labeled by the first author, so to address this issue, we worked with multiple annotators and performed lyrics labeling through majority voting, so we will introduce this procedure in the next section.

B. Majority Voting

First, we provided 240 lyrics (without emotional IDs) to six male members of my laboratory. Subsequently, they were asked to verify six categories and provide detailed description for each classification.

Next, as show in Fig. 2(a), the annotators were asked to review the instruction document. Using the example lyrics provided below the instructions as reference, they were instructed to assign a single ID (from 1 to 6) to each of the 240 lyrics. Even in cases where multiple categories seemed applicable, annotators were asked to intuitively select the ID of the most closely matching example lyrics. After the annotation process was completed by the 6 annotators, we determined the true emotion label for each lyric through majority voting. For this experiment, we only included lyrics where 4 or more annotators agreed on the same emotion ID. We excluded cases where multiple emotions received an equal number of votes. The resulting gold standard dataset after majority voting is shown in Fig. 2(b).

IV. Lyrics Emotion Classification by an LLM

This section describes how to provide prompts for emotion classification of lyrics using an LLM, as well as various evaluation results. The models used for emotion classification are GPT-4o (*gpt-4o-2024-08-06*).

[3]https://open.spotify.com/

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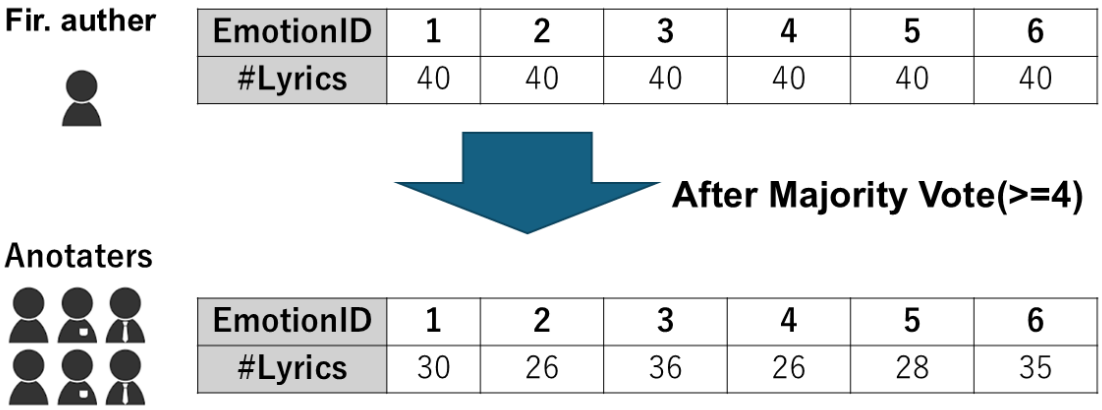
Instraction to anotaters

The example lyrics are numbered from <Emo1> to <Emo6>. Please assign a number to the test lyric that corresponds to the most similar example lyrics. Even if you feel multiple numbers might apply, please intuitively select the single number that best represents the closest match among the example lyrics.

Example Lyrics

<Emo1>: We're always the same ...(example lyric 1), <Emo1>: Mom's kindness and Dad's crybaby ...(example lyric 2)
<Emo2>: There may be days when ...(example lyric 1), <Emo2>: Riding on a lively beat...(example lyric 2)
<Emo3>: The days and tears spent ...(example lyric 1), <Emo3>: Beyond the night, through the ...(example lyric 2)
<Emo4>: When I met you, my days...(example lyric 1), <Emo4>: The miracle of meeting you...(example lyric 2)
<Emo5>: Bidding farewell as you...(example lyric 1), <Emo5>: Please, let me stay by your side...(example lyric 2)
<Emo6>: While feeling that I cannot...(example lyric 1), <Emo6>: Confessing one's true...(example lyric 2)

(a) Instructions and Example Lyrics for Annotation Task



(b) Dataset Balance Before and After Majority Voting

Fig. 2: Annotation Process Flow

A. Prompt for Emotion Classification of Lyrics

We discusses the prompt design for emotion classification of lyrics using GPT-4o. Fig. 4 shows specific examples of prompts. As the input, lyrics of the candidate emotion classes are provided as few-shots as well as the test lyric whose emotion to be classified, and GPT-4o is asked to answer the number of the candidate emotion class as the prediction result. We adopted this method because if the class names of the candidate emotion classes are directly given, GPT-4o would interpret them. Therefore, the class names are indicated by numbers, and the definitions of candidate emotion classes are given through few-shot lyrics rather than the natural language descriptions.

1) Presenting the Test Lyric and Few-Shot Lyrics: Next, we explain how to present the test lyric and few-shot lyrics given to the prompts created in the previous section. As for the test lyric, we first assume that providing only the most informative excerpted parts instead of providing the entire lyric would improve performance compared to providing the entire lyric. Based on this assumption as well as the length distribution[4] of all the lyrics in the dataset, we performed a preliminary evaluation with a held-out dataset other than those presented in section III-A, consisting of 36 lyrics. With this preliminary evaluation, we decided to manually extract the most informative approximately 250 characters from each lyric[5], and perform the evaluation of emotion classification when providing GPT-4o with the manually extracted part of the test lyric, where we compare its performance with the case when providing GPT-4o with the entire lyric.

Next, as the few-shots, two lyrics are selected for each

[4]Minimum 157 characters, maximum 1,604 characters, and average 556 characters.

[5]We evaluate the cases where providing GPT-4o with the manually extracted most informative 50, 100, 150, $\cdots$, 600, 650, and 700 characters, where we obtained the best performance with 250 characters.

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Prompt for Summarization
You are an excellent lyrics summarization system.
Please read the lyrics in [Input Data], extract parts related to central themes, impressive expressions, and scenic descriptions, and modify them to an optimal character length.
Do not output any text other than [Output Format]. Here are examples of input and output below.

[Input Data] = lyric
[Output Format] = optimal text

Fig. 3: Prompts for Optimal Lyrics Summarization

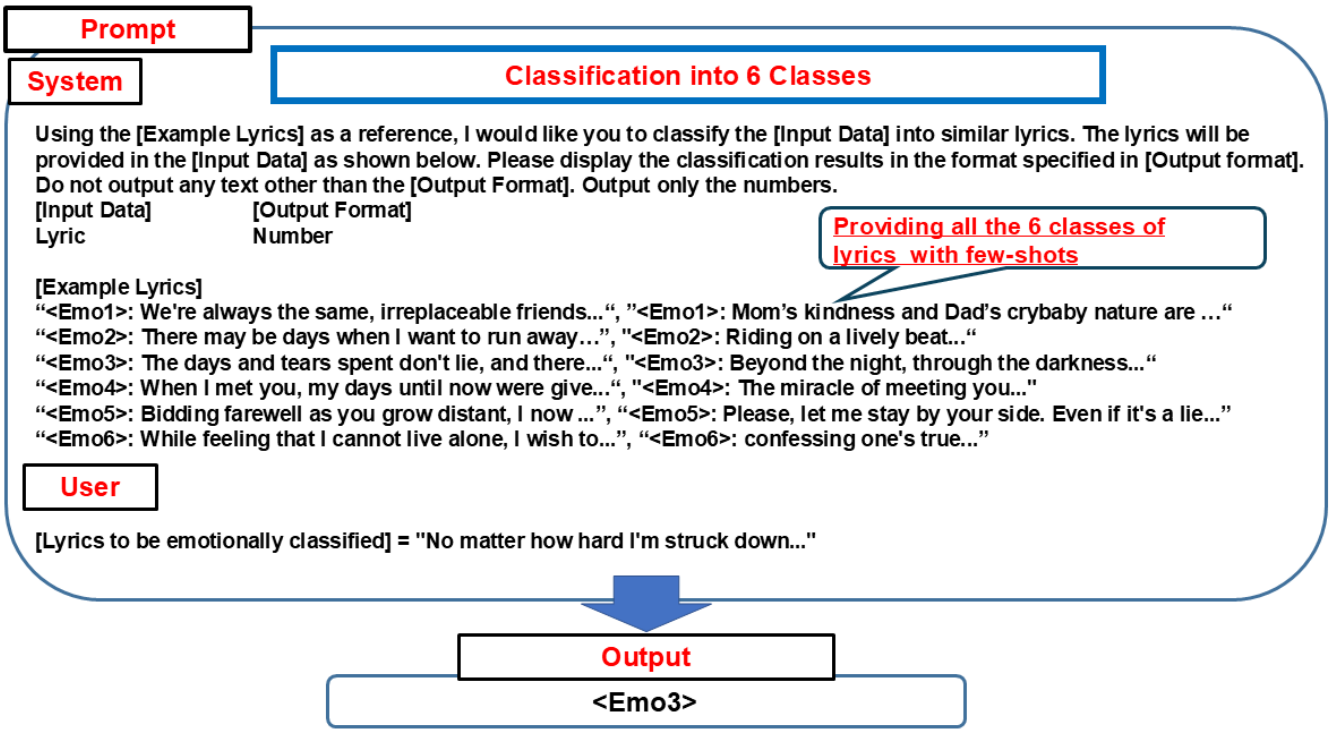

Fig. 4: Prompts (with two lyrics as few-shots for each emotion class) and Output Examples by GPT-4o

emotion class. Here, following the above result of optimizing the length of the test lyric, instead of providing the entire lyric, parts (about 250 characters) that are considered to strongly express the emotion of the lyric are manually selected and excerpted.

2) Extractive Summarization of Lyrics into Optimal Length: In this section, we evaluate the emotion classification performance when GPT-4o performs the task of extractive summarization of the most emotionally expressive parts of lyrics, a task that was previously done manually, and experiments are conducted using the resulting lyric data. In this experiment, we used two methods: asking GPT-4o to summarize lyrics in 250 characters and to summarize them in the optimal length. The prompt given to GPT-4o for summarizing the lyrics is as shown in Fig. 3. Using this method, all lyric data were processed by GPT-4o to create dataset with 250 characters and another with the optimal length. The distribution of the lyrics is shown in Fig. 5. The distribution of our original lyrics is shown in Fig 5(a). On average, the original lyrics were 556 characters, but some longer ones exceeded 1600 characters, while there were also short lyrics with less than 200 characters. The distribution when we instructed GPT-4o to limit lyrics to 250 characters is shown in Fig. 5(b). While GPT-4o didn't strictly adhere to this character limit, we can

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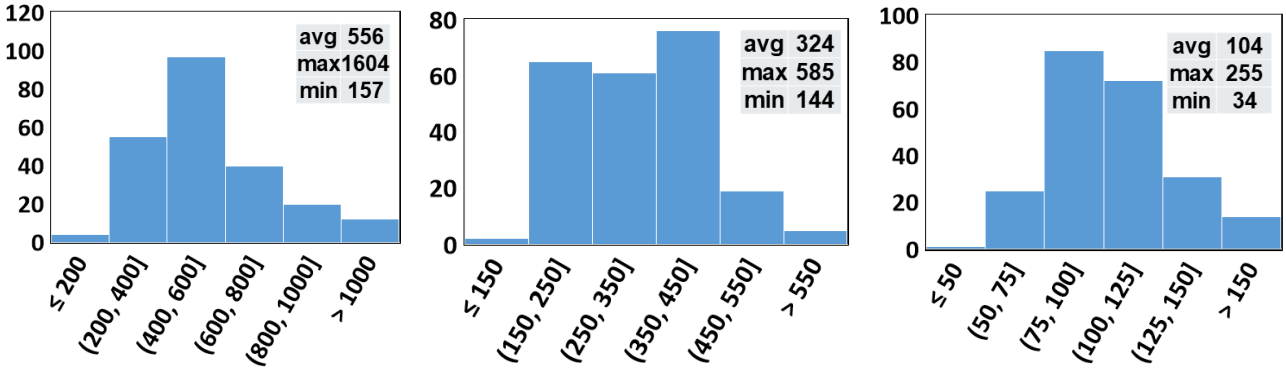

(a) Before summarization (b) Summarized into 250 characters (c) Summarized into optimal

Fig. 5: Characters length distribution of lyrics

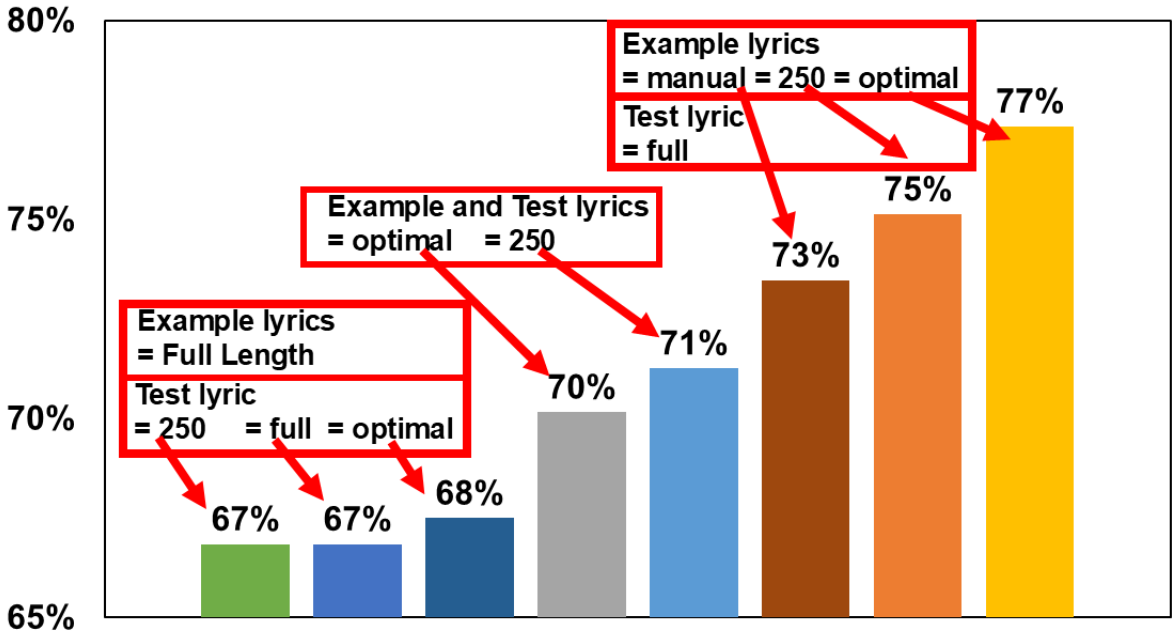

Fig. 6: Evaluation Results (total accuracy of classification into 6 classes)

see a significant reduction in length compared to the original data. The distribution when we asked GPT-4 to optimize the lyrics is shown in Fig 5(c). Most optimized lyrics fall between 75 and 125 characters, demonstrating a substantial reduction in length.

B. Emotion Classification of Lyrics

1) Classifications into 6 classes: This section evaluates the 6 class classification. The results is shown in Fig. 6. First examine the three bar graphs on the right. In these experimental settings, the test lyrics were given the full text. For the example lyrics, from right to left, we have: lyrics optimally summarized by GPT-4o, lyrics specified to be 250 characters by GPT-4o, and lyrics manually excerpted to be 250 characters. As shown in the bar graphs, using lyrics summarized by GPT-4o achieved higher accuracy than using manually excerpted lyrics that focused on characteristic sections. While manually excerpting or summarizing lyrics requires considerable time, the ability to easily generate effective summaries using GPT-4o represents a significant advantage. Furthermore, regarding lyrics summarization, we found that allowing GPT-4o to determine the optimal length independently performed better than imposing character count restrictions to maintain a certain length.We think this is because when character limits are

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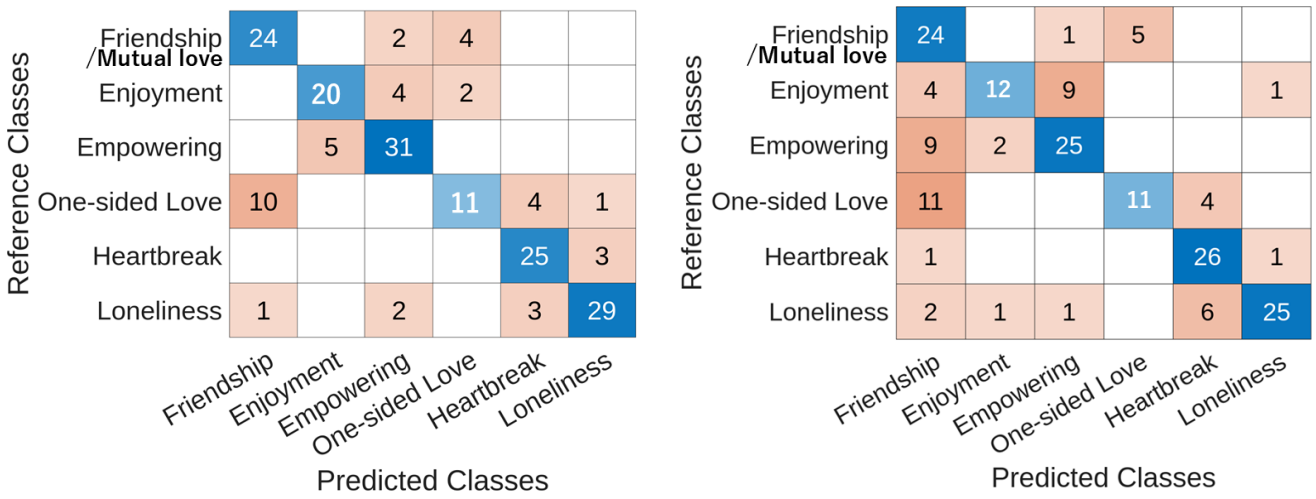

(a) Optimal length for Example lyrics, Full length for test lyrics

(b) Full Length for Example lyrics, optimal length for test lyrics

Fig. 7: Confusion Matrices of Emotion Classification

imposed on lyrics summarization, essential features of the lyrics may be removed, or unnecessary content may be added to meet the length requirement.

Next, examining the middle bar graph (which shows results from using 250 character summarized lyrics for both example and test lyrics and using optimally-length summarized lyrics for both).Unlike the three graphs on the right where only example lyrics were summarized, performance deteriorated when summarized lyrics were used for both test and example lyrics.

Finally, let's examine the three bar graphs on the left. In these experiments, example lyrics were given the full text, while test lyrics were given (from right to left): GPT-4o optimally summarized lyrics, full lyrics text, and GPT-4o summarized lyrics constrained to 250 characters. These combinations yielded the poorest performance among all configurations. These experiments revealed that for optimal GPT-4o performance, example lyrics should not use the full text but rather contain characteristic features, while test lyrics should use the complete text rather than shortened versions.

2) Confusion Matrix: In this section, we analyze the detailed classification results using a confusion matrix. Fig. 7(a) shows the confusion matrix for the best-performing configuration, where example lyrics were optimally summarized by GPT-4O and test lyrics were given the full text.The reference classifications are shown on the left side, and the predicted classifications are shown at the bottom. The results indicate that lyrics about "Empowering" were most accurately classified, while lyrics about "One-sided love" had the highest misclassification rate.

Further analyzing misclassification patterns, we found that "Hreatbreak" and "Loneliness" lyrics were frequently misclassified as each other. This can be attributed to the fact that these two classes share a stronger negative emotional element compared to the other four classes.

Furthermore, "Enjoyment" and "Empowering" lyrics showed mutual misclassification tendencies. While these categories differ conceptually—"Enjoyment" depicting characters actively experiencing pleasure versus "Empowering" expressing motivation for oneself or others—both share high-energy, vibrant characteristics in their expression. We hypothesize that this tonal similarity made it challenging for GPT-4o to establish clear distinctions between these categories.

While "One-sided love" had the lowest accuracy rate, Fig. 7(a) shows many incorrect classifications clustered in "Friendship/mutual love". This appears to be due to the difficulty in distinguishing between content types. The distinction between "Friendship/mutual love" and "One-sided love" is that lyrics about family, friends, or partners fall under "Friendship/mutual love", while lyrics about affection for others fall under "One-sided love". This indicates the need to develop prompts that can better help differentiate between these categories.

Fig. 7(b) presents results when full lyrics were used for example lyrics and GPT-4 optimally summarized lyrics for test lyrics. Compared to Fig. 7(a), merely changing the length configuration of example and test lyrics produced significant differences in classification performance. All categories except "Friendship/mutual love" and "Heartbreak" showed decreased accuracy. Notably, misclassifications into the "friendship/mutual love" category increased substantially, with 4 instances from "Enjoyment" and 9 from "Empowering" being incorrectly classified as "Friendship/mutual love". This pattern is understandable given that enjoyable situations often involve friends, and courage-related content may involve encouraging friends. Our findings suggest that careful attention to the length

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configuration of example and test lyrics can effectively prevent such subtle thematic misclassifications.

V. Conclusion

In this study, we investigated methods for adapting GPT-4o to Japanese lyrics emotion classification. We developed a dataset of 181 lyrics categorized into six classifications to evaluate GPT-4o's performance. Our developed prompts enabled GPT-4O to achieve over 75% accuracy in six-class classification. We found optimal performance when using GPT-4o summarized lyrics for examples and full lyrics for test, rather than uniformly using full lyrics. Future work should focus on developing prompts to better differentiate between "Friendship/mutual love" and "Heartbreak" categories. Additionally, increasing the number of annotators would enhance the validity of our majority-vote gold standard data. Further research should also examine classification performance using other open and closed LLMs.

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